VB.NET PROGRAMMING WORKSHEET

UNDERSTANDING FUNCTIONS

A-Level Computer Science | Beginner to Intermediate

About this worksheet

This worksheet introduces functions step by step. Learners begin by declaring and calling simple functions, then progress to parameters, return types, calculations, selection, loops and larger programs that combine several functions.

Learning outcomes

Explain what a function is and why functions are useful.
Declare, call and test functions in VB.NET.
Pass arguments to functions using parameters.
Choose suitable return types and use Return to send values back.
Use selection and repetition inside functions.
Combine functions to solve practical programming problems.

Before you begin: function or Sub procedure?

A Function is a named block of code that performs a task and returns a value. A Sub procedure performs an operation but does not return a value.

```
Function FunctionName() As Integer
    'Statements to perform a task
    Return 10
End Function
```

Example: the function below returns the integer 25. Main calls the function and displays the returned value.

```
Function GetNumber() As Integer
    Return 25
End Function

Sub Main()
    Console.WriteLine(GetNumber())
End Sub
```

Expected output: 25

LESSON 1: Creating Simple Functions

Focus: Understanding function declaration, function calls and return values.

Objective: Create and call functions that return values without receiving parameters.

Exercise 1 – Returning a Number

Create a function called **`GetAge`**. The function must return the number 18. In the Main procedure, call the function and display the returned value.

1. What keyword is used to declare a function in VB.NET?
2. Which keyword is used to send a value back from a function?
3. What will happen if the function is called but its returned value is not displayed or stored?

Exercise 2 – Returning a Name

Create a function called **`GetStudentName`**. The function must return the name "John". Call the function and display the returned name.

Exercise 3 – Returning a School Name

Create a function called **`GetSchoolName.`** The function must return the name of your school. Display the returned value using a **`Console.WriteLine()`** statement.

Exercise 4 – Returning a Fixed Calculation

Create a function called **`CalculateSum.`** The function must declare two local variables named num1 and num2, assign them the values 20 and 30, calculate their sum, and return the answer. Display the returned answer in Main.

Reflection: What is the difference between displaying a result inside a function and returning a result from a function?